Indian Institute of Technology Hyderabad
Signal Processing, Fall 2026
Quiz 0, 06.08.2026, 10 points

1. Define a *discrete-time signal*. Give two examples of natively discrete-time signals. (2)
2. Define a *discrete-time system*. Give two examples of commonly occurring discrete-time systems. (2)
3. A discrete-time system computes the average of the *current* sample and the *sample before its previous* sample. What type of a filter is this system? Justify your answer. (2)
4. Show that the basis vectors of the N-point DFT are orthogonal. (2)
5. Given a signal $\mathbf{x} = [2, 0, 2, 0, 2, 0, 2, 0]^T$. Find its 8-point DFT without actually carrying out the DFT computation. Justify your answer. (2)